

Forecasting: Fundamentals, Role in Physics and Physics-Informed Machine Learning

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Forecasting refers to the art and science of making predictions about the future using information from the past and present. From planning grocery orders to anticipating the trajectory of a planet, successful forecasts underpin decision making across science, engineering and economics. This report introduces the fundamentals of forecasting and time-series modelling, explains how classical and machine-learning methods are used to predict the evolution of physical systems, and then describes the emerging field of *physics-informed neural networks* (PINNs), where machine learning models incorporate physical laws. Throughout, explanations draw inspiration from Richard Feynman's intuitive style: complex techniques are unpacked using thought experiments and analogies.

Introduction: What Is Forecasting?

Imagine watching a ball thrown into the air. You can predict where it will land in two broad ways. One is by remembering previous throws; the other is by using the law of gravity and the equations of motion. The first is pattern recognition, while the second is physics. *Forecasting* combines these perspectives. In its broadest sense, forecasting is the process of making statements about events whose actual outcomes have not yet been observed. It makes use of past and present information and always involves risk and uncertainty

【236179935959006†L174-L199】. Unlike a budget, which is a static plan, a forecast is a dynamic projection that can be revised as new information arrives 【919849784953045†L435-L469】. The selection of forecasting methods depends on the problem, the available data, and the cost of errors 【236179935959006†L206-L247】.

The basic structure of a forecast consists of three elements: (1) a set of assumptions about how the system works (this may range from physical laws to qualitative judgements), (2) data representing observations of the system, and (3) a computational method that combines the assumptions and data to produce predictions. Classically, forecasts are divided into *qualitative* and *quantitative* methods 【236179935959006†L206-L247】. Qualitative methods rely on expert opinion, analogies and market research, while quantitative methods use statistical models and machine learning to extrapolate patterns from data.

Time-Series Forecasting

Basic concepts

A time series is a sequence of observations ordered in time, such as daily temperatures, stock prices, or the position of a pendulum. Time-series analysis examines such sequences to understand underlying structure and predict future values. The key distinction from cross-sectional data is that the order of observations matters; the techniques exploit the dependency of current values on past values 【420234280771974†L210-L229】. In Feynman's

words, it is like watching a series of frames in a film: you cannot rearrange the frames without changing the story.

Time-series data often display patterns such as trend (long-term increase or decrease), seasonality (regular periodic fluctuations), and randomness. Forecasting methods must account for these components. Most quantitative methods assume the series can be decomposed into deterministic (trend and seasonality) and stochastic (random) parts, and they attempt to model or filter each part separately.

Classical statistical models

One of the most widely used families of statistical forecasting models is the *autoregressive integrated moving average* (ARIMA) model. An $ARIMA(p, d, q)$ process models the current value of a time series as a combination of (i) p past values (the autoregressive part), (ii) q past forecast errors (the moving average part), and (iii) differences of order d to achieve stationarity (the integrated part). ARIMA models are popular because they capture both autoregressive and moving-average dynamics while handling non-stationarity through differencing

【140663421906524†L76-L82】. For example, to forecast the position of a swinging pendulum, one might note that the angle at time t depends on its values at several previous times and on the cumulative effects of random disturbances; differencing removes slow drifts.

Other classical methods include exponential smoothing, state-space models, and spectral analysis. They all share the principle of parsimony: explain the data using as few parameters as possible. The strength of these models is their transparency; their parameters often have physical interpretations and can be estimated from relatively small datasets.

Machine-learning models for sequential data

When patterns are highly nonlinear or high-dimensional, classical models struggle. Machine-learning models, particularly recurrent neural networks (RNNs) and long short-term memory (LSTM) networks, offer flexible alternatives. An RNN processes a sequence one element at a time and stores a representation of the past in its hidden state. However, standard RNNs suffer from vanishing gradients that make it hard to learn long-range dependencies. LSTM networks address this by introducing a memory cell with gates controlling what information to retain or forget 【255653207246584†L67-L74】. This architecture allows LSTMs to learn long-term temporal relationships and has made them a staple for tasks such as speech recognition, translation, and time-series forecasting 【255653207246584†L112-L113】 【255653207246584†L217-L219】.

From a Feynman perspective, an LSTM can be thought of as a clever notebook: at each time step it decides which facts from previous pages to keep and which to erase, enabling it to make predictions that depend on context far back in time. These models are powerful pattern recognizers; they learn from data without explicit knowledge of the underlying physics. This strength is also a weakness: they require large datasets and often behave like black boxes, making their predictions hard to interpret.

Forecasting in Physics

Historical context

Physics has long been concerned with forecasting. The earliest known weather predictions were pattern-based: farmers and sailors relied on recurring natural signs. In the early twentieth century, Vilhelm Bjerknes and Lewis Fry Richardson pioneered *numerical weather prediction*, applying the Navier–Stokes equations and thermodynamics to forecast atmospheric fields. These physics-based models became the backbone of modern meteorology when computers allowed their solution. However, recent advances in machine learning have returned forecasting to its pattern-based roots. The European Centre for Medium-Range Weather Forecasts has demonstrated an AI-based forecasting system that can sometimes outperform traditional models while using orders of magnitude less computational energy [377605596713146†L34-L66]. Yet this AI still relies on data generated by physics-based simulations, illustrating the complementarity of data and theory [377605596713146†L34-L66].

Physical and data-driven forecasting

In physics, predictions are typically derived from fundamental laws—Newton’s equations, Maxwell’s equations, Schrödinger’s equation—supplemented by initial and boundary conditions. These differential equations describe how a system evolves exactly; if we know the initial state and parameters, we can compute the future state precisely. But in practice we rarely have perfect information; measurement noise, unresolved scales and model approximations introduce uncertainty. Therefore physicists have long used statistical methods to quantify errors and propagate them forward. In weather forecasting, for example, ensembles of numerical simulations with slightly different initial conditions are run to estimate the spread of possible outcomes.

Machine learning offers a complementary approach: instead of solving the governing equations, we fit flexible functions to observed data. This is like inferring the trajectory of a ball from seeing many throws without ever deriving Newton’s laws. Modern AI models can capture subtle patterns from vast datasets, revealing empirical relationships that may be hidden in the equations [377605596713146†L125-L129]. However, purely data-driven models may fail when encountering situations unlike those in the training data and can lack physical consistency (for example, predicting negative rainfall). The ideal forecast blends physical understanding with data-driven learning.

Physics-Informed Neural Networks (PINNs)

Motivation and basic formulation

Physics-informed neural networks (PINNs) attempt to combine the strengths of mechanistic and machine-learning models. The key idea is to embed known physical laws—usually expressed as partial differential equations (PDEs) or ordinary differential equations (ODEs)—directly into the loss function used to train a neural network. In addition to minimizing the

discrepancy between the network's predictions and available data (the data loss), the training procedure penalizes violations of the governing equations (the physics loss) [580875199186709†L242-L260]. This can be written schematically as

$$L(\theta) = \sum_{i=1}^N \|u_{\theta}(t_i) - y_i\|^2 + \lambda \sum_{j=1}^M \|F[u_{\theta}](x_j)\|^2,$$

where u_{θ} is the neural network parameterized by weights θ , y_i are the observed data, $F[u_{\theta}]$ denotes the PDE residual, and λ balances the two terms.

This formulation encourages the network to satisfy the governing equations everywhere in the domain, even where data are scarce or noisy [580875199186709†L140-L155]. The physics loss acts as a regularizer, increasing generalization and reducing the amount of training data needed [580875199186709†L140-L155]. In Feynman's language, the network is not merely memorizing; it is being taught that "the answer must also obey the laws of nature," much like a student who is told the result must also satisfy energy conservation.

Benefits and limitations

PINNs provide several advantages over purely data-driven neural networks. First, they embed prior knowledge, so they can make accurate predictions in regimes where data are scarce or noisy [182676526810939†L118-L133]. Second, PINNs are mesh-free: they approximate the solution of PDEs without requiring a discretized grid, which can be advantageous for high-dimensional problems [580875199186709†L186-L194]. Third, they can solve *inverse problems*, where one wants to infer unknown parameters or boundary conditions from partial observations [182676526810939†L118-L133]. Fourth, by combining data and physics, PINNs can handle ill-posed problems and incorporate sparse measurements. [182676526810939†L118-L133]

Nevertheless, the approach has limitations. The convergence theory of PINNs is still immature [182676526810939†L135-L143]. There is no unified training strategy; hyper-parameter choices, sampling of collocation points and scaling of loss terms can significantly affect performance [182676526810939†L135-L143]. Calculating high-order derivatives through automatic differentiation can be computationally expensive, especially for complex PDEs [182676526810939†L135-L143]. Moreover, PINNs may struggle to capture high-frequency or multiscale solutions [182676526810939†L135-L143]. As Feynman might put it, embedding the laws of nature does not guarantee that the network will discover every subtlety of the solution; the network is still only as good as the complexity it can represent and the guidance it is given.

Applications

PINNs have been successfully applied across many areas of physics and engineering. They have been used to model fluid flows, electromagnetism, structural mechanics, optical waveguides, seismic waves, and biological systems [25828375903107†L197-L303]. In cases where obtaining accurate numerical solutions is expensive, PINNs can serve as surrogate models for rapid evaluation or as components of control systems. For instance, they have been used for model predictive control of dynamical systems, where the physics loss enforces the dynamics while the data loss guides the control objective [580875199186709†L263-L266]. Their ability

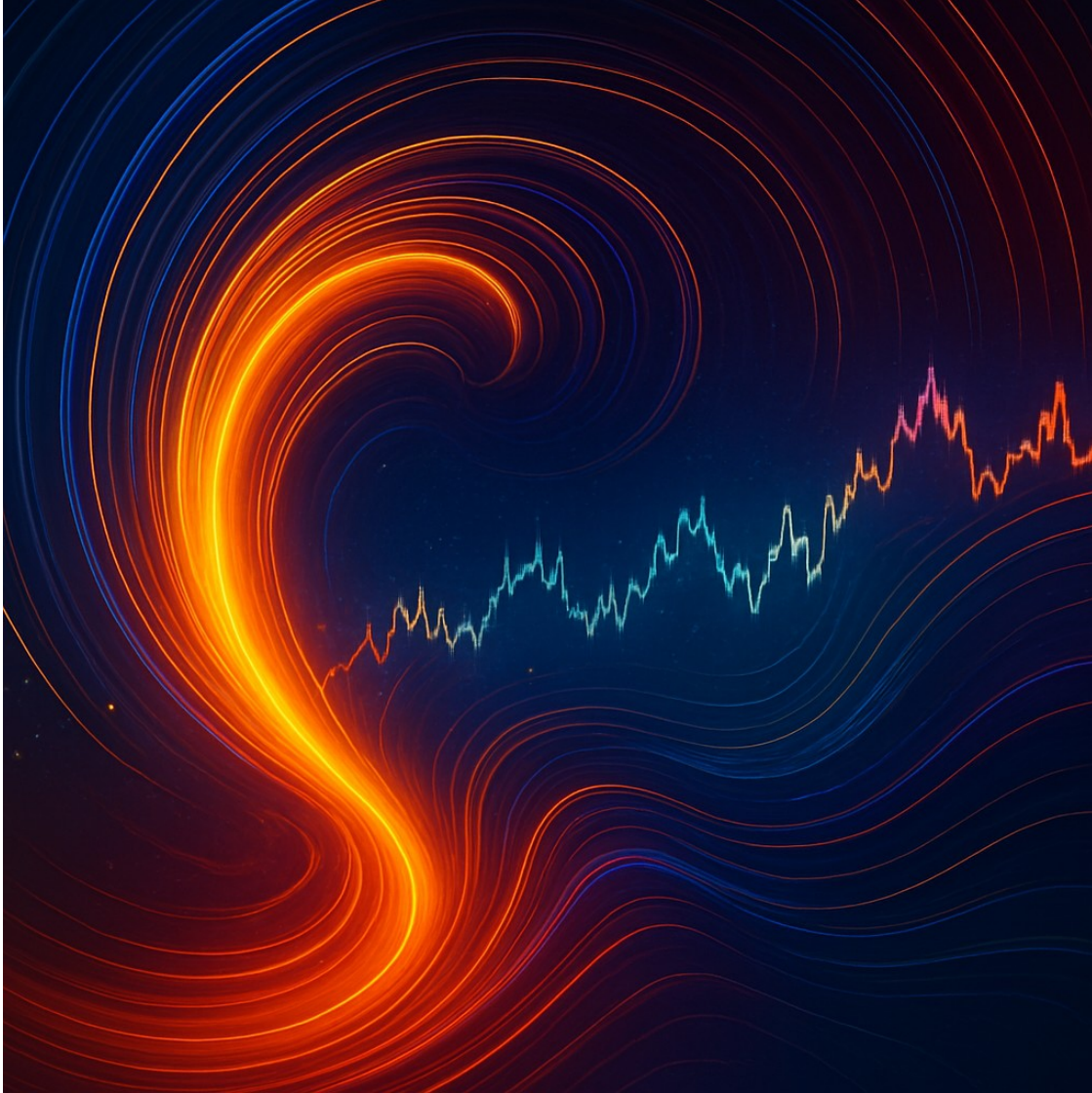
to handle inverse problems makes them attractive for tasks like inferring material properties from limited measurements, a common challenge in geophysics and medical imaging.

Machine Learning and Physics: A Symbiotic Future

Forecasting lies at the intersection of mathematics, statistics, physics and computer science. Classical models offer interpretability and consistency with known laws, while machine learning excels at pattern recognition and high-dimensional function approximation. The future of forecasting will likely involve a symbiotic relationship where machine learning algorithms are informed by physics and vice versa. AI-based weather forecasting shows that data-driven models can sometimes achieve better accuracy than state-of-the-art physics models while using a fraction of the computational resources [377605596713146†L34-L66]. At the same time, these AI systems depend on high-quality data generated by physics-based simulations and observations, and they may fail under conditions not represented in the training data. PINNs represent a promising step towards unifying these approaches by embedding physical constraints into neural networks.

Conclusion

Forecasting is not magic; it is an exercise in extracting patterns and applying knowledge to make educated guesses about the future. Whether we are predicting tomorrow's weather, the future price of copper, or the trajectory of an asteroid, the same principles apply: combine the best available information with the best available theory. Classical statistical models like ARIMA provide transparent and parsimonious descriptions of time-series data, while modern machine-learning models like LSTMs allow us to capture complex nonlinear patterns. Physics-based forecasting uses laws of nature to predict the future; data-driven forecasting learns patterns from experience. Physics-informed neural networks offer a way to blend these philosophies, enforcing physical consistency while leveraging the flexibility of neural networks. In the spirit of Richard Feynman, understanding comes from learning the rules and then checking that our predictions obey them; forecasting is simply the application of that philosophy to the future.



An abstract visualization of forecasting in physics, illustrating swirling patterns and a time-series trace. The interplay of order (smooth curves) and randomness (fluctuations) reflects the dual role of deterministic laws and stochastic influences in prediction.